EXPERIMENT 1

MAKE YOUR OWN PARALLEL AND SERIES CIRCUITS USING BREADBOARD FOR ANY APPLICATION OF YOUR CHOICE

VIVA QUESTIONS & ANSWERS

1. **What is the difference between the series and parallel circuits?**

Answer: In a series circuit, current is the same through all elements, but in a parallel circuit, voltage is the same across all branches.

2. **What happens to the total resistance in a series and parallel circuits when you add more resistors?**

Answer: The total resistance increases because resistors add up in a series circuit, while in the parallel circuit the total resistance decreases, because the reciprocal of the total is the sum of reciprocals.

3. **What is a breadboard?**

Answer: A breadboard is a tool for building and testing circuits without soldering. It has interconnected rows and columns for placing electronic components.

4. **How do you ensure proper connections in a breadboard circuit?**

Answer: By carefully placing components in the correct rows and columns, and ensuring that wires and terminals are not loose.

5. **Why do LEDs need resistors in a circuit?**

Answer: To limit current through the LED and prevent it from burning out.

6. **How can you increase the brightness of LEDs in a series circuit.**

Answer: By increasing the supply voltage or reducing resistor values, ensuring it's within safe limits for LEDs.

7. **Name the applications of series and parallel circuits**

Answer: Series circuits are used in Christmas lights, where all bulbs are in series and controlled by a single switch. Parallel circuits are used in home wiring, where each appliance gets the same voltage and can be operated independently.

8. **Which is better for power efficiency, series or parallel? Why?**

Answer: Parallel circuits are better, as they offer constant voltage and independent operation of devices.

9. Can a switch be used to control individual LEDs in a parallel circuit?

Answer: Yes, placing a separate switch in each branch allows individual control of LEDs.

10. What precautions must be taken while building circuits on a breadboard?

Answer:

1. Check all connections twice
2. Use resistors to protect components
3. Avoid short circuits
4. Ensure power supply polarity is correct

EXPERIMENT 2

DEMONSTRATE A TRAFFIC LIGHT CIRCUIT USING BREADBOARD

VIVA QUESTIONS

1. **What is the role of LEDs in this experiment?**

Answer: LEDs are used to visually simulate the functioning of a traffic light, indicating signals like Stop (Red), Ready (Yellow), and Go (Green).

2. **Why do we need resistors in an LED circuit?**

Answer: Resistors limit the current flowing through each LED, protecting them from burning due to high voltage.

3. **What is the purpose of the push buttons in this setup?**

Answer: Push buttons are used to manually switch ON or OFF the LEDs, simulating the control of traffic signals.

4. **Why is a 9V battery used in this experiment?**

Answer: A 9V battery provides a portable power source to drive the LED circuit. The voltage is dropped across resistors to safely power the LEDs.

5. **How does a push button work in a circuit?**

Answer: A push button is a momentary switch that completes or breaks the circuit when pressed, allowing or stopping current flow to the LED.

6. **How is the breadboard useful in this experiment?**

Answer: The breadboard allows us to connect components without soldering, making it easy to modify and test the circuit.

7. **What safety precautions must be followed while using a 9V battery with LEDs?**

Answer: Always use appropriate resistors to prevent overcurrent, and ensure correct polarity to avoid damaging the LEDs.

8. **What happens if you connect an LED without a resistor to a 9V battery?**

Answer: The LED may burn out immediately due to excessive current.

9. **What is the function of each LED color in this traffic light circuit?**

Answer: Red LED – Stop, Yellow LED – Caution or Ready, and Green LED – Go.

10. **Which type of circuit is a traffic light—series or parallel?**

Answer: A traffic light circuit is a parallel circuit, so that each LED can be controlled independently without affecting the others.

EXPERIMENT 3

BUILD AND DEMONSTRATE AUTOMATIC STREET LIGHT USING LDR

VIVA QUESTIONS

1. **What is the main purpose of using an LDR in this experiment?**
   Answer: The LDR is used to detect the ambient light levels. It acts as a sensor that triggers the LED to turn ON when it becomes dark and OFF when it's bright.
2. **How does the resistance of an LDR change with light?**
   Answer: The resistance of an LDR decreases when the light intensity increases, and it increases when the light intensity decreases.
3. **Why is a resistor connected in series with the LED?**
   Answer: A resistor is used in series with the LED to limit the current passing through it and prevent the LED from burning out due to excess current.
4. **What happens to the LED when the LDR is exposed to darkness?**
   Answer: In darkness, the LDR's resistance increases, changing the voltage across the LED circuit and allowing current to flow through the LED, thereby turning it ON.
5. **What type of circuit configuration is used to connect the LDR and resistor?**
   Answer: A voltage divider circuit configuration is used, where the LDR and resistor are connected in series to divide the voltage based on light intensity.
6. **Can LDRs detect artificial light as well as sunlight?**
   Answer: Yes, LDRs can detect any form of visible light, including artificial light from bulbs and natural light from the sun.
7. **What is the role of the breadboard in this experiment?**
   Answer: The breadboard allows us to connect the electronic components temporarily without soldering, making it easier to build and modify the circuit.
8. **What would happen if the LDR and resistor positions are swapped in the voltage divider?**
   Answer: Swapping the LDR and resistor will reverse the voltage behavior, possibly causing the LED to turn ON in light and OFF in dark, which is the opposite of the intended behavior.
9. **Why is a 9V battery used in this circuit?**
   Answer: A 9V battery provides a sufficient voltage supply for the LED and other components, and it is commonly available and easy to connect to a breadboard circuit.
10. **Which type of circuit is used in this experiment – series or parallel?**
    Answer: The circuit mainly follows a series configuration in the voltage divider part, but the overall circuit can be viewed as a combination, depending on how the LED is connected relative to the sensor and resistor.

EXPERIMENT 4

SIMULATE THE ARDUINO LED BLINKING ACTIVITY IN TINKERCAD

VIVA QUESTIONS

1. **What is the purpose of the `pinMode()` function in Arduino?**

Answer: It sets a specific pin to behave as either an input or an output.

2. **What does the `digitalWrite()` function do?**

Answer: It sends a HIGH or LOW signal to a digital pin to turn devices ON or OFF.

3. **Why do we use a resistor with an LED?**

Answer: To limit the current passing through the LED and prevent it from burning out.

4. **What is the value of the resistor used in the blinking LED circuit? Why?**

Answer: 220 ohms; it's a safe value to limit current and protect the LED.

5. **What is the `delay()` function used for in Arduino?**

Answer: It pauses the program for a specified time in milliseconds.

6. **What does the Arduino UNO pin 13 have that makes it suitable for this experiment?**

Answer: Pin 13 is connected to an onboard LED, useful for basic testing and blinking.

7. **How does Tinkercad help in learning Arduino?**

Answer: It allows simulation of Arduino projects without physical hardware, making testing and learning easy.

8. **What will happen if you remove the resistor from the LED circuit?**

Answer: The LED might receive too much current and get damaged or burn out.

9. **Can you make the LED blink faster? How?**

Answer: Yes, by reducing the delay time in the delay() function (e.g., to 200 ms).

10. **What is the basic difference between analog and digital pins in Arduino?**

Answer: Digital pins can be HIGH or LOW (0 or 1), while analog pins can read a range of voltage values using ADC.

EXPERIMENT 5

BUILD AND DEMONSTRATE AN ARDUINO LED BLINKING ACTIVITY USING ARDUINO IDE

VIVA QUESTIONS

1. **What is the use of the `setup()` function in Arduino?**

Answer: The `setup()` function runs once at the beginning and is used to initialize pin modes, communication settings, etc.

2. **What does `pinMode(13, OUTPUT)` mean?**

Answer: It sets pin 13 as an output so it can send voltage to components like an LED.

3. **Why do we use a resistor with an LED?**

Answer: To limit the current flowing through the LED and prevent it from burning out.

4. **What does the `digitalWrite()` function do?**

Answer: It sets a digital pin to either HIGH (5V) or LOW (0V).

5. **What is the function of the `delay()` function?**

Answer: It pauses the program for a specified time in milliseconds.

6. **Can you blink an LED using a pin other than 13?**

Answer: Yes, any digital pin can be used; the code and circuit must be updated accordingly.

7. **What is the voltage output from a HIGH signal in Arduino?**

Answer: Approximately 5 volts on most Arduino boards.

8. **What are the two main sections in every Arduino program?**

Answer: `setup()` for initialization and `loop()` for continuous execution.

9. **What will happen if you connect the LED directly to 5V and GND without a resistor?**

Answer: The LED may burn out due to excess current.

10. **How do you upload a program to the Arduino board?**

Answer: By connecting the board via USB, selecting the correct board and port in the Arduino IDE, and clicking the Upload button.

EXPERIMENT 6

INTERFACING IR SENSOR AND SERVO MOTOR WITH ARDUINO

VIVA QUESTIONS

1. **What is the function of an IR sensor in this experiment?**

Answer: The IR sensor detects the presence of an object by emitting infrared light and sensing its reflection. When an object is detected, it sends a digital signal to the Arduino.

2. **Why do we use a servo motor instead of a DC motor in this application?**

Answer: A servo motor allows precise control of angular position, which is required here to rotate the motor to a specific angle (like 90°) when the IR sensor detects an object.

3. **Which Arduino pin type is used to control a servo motor?**

Answer: Servo motors are controlled using PWM (Pulse Width Modulation) pins on the Arduino.

4. **What is the purpose of the `Servo.h` library in Arduino?**

Answer: The `Servo.h` library provides built-in functions to control servo motors easily, such as setting their angle using `write()` and attaching them to specific pins using `attach()`.

5. **Why is the IR sensor's OUT pin connected to a digital pin on Arduino?**

Answer: The IR sensor's OUT pin gives a digital HIGH or LOW signal depending on object detection, which can be directly read by an Arduino digital input pin.

6. **What voltage supply is required for the IR sensor in this circuit?**

Answer: The IR sensor requires a 5V DC supply from the Arduino for proper operation.

7. **In the code, what does `digitalRead(sensorpin)` do?**

Answer: `digitalRead(sensorpin)` reads the HIGH (1) or LOW (0) signal from the IR sensor to determine whether an object is present.

8. **What will happen if the IR sensor detects an object in this program?**

Answer: If the IR sensor detects an object, the servo motor rotates to 90°, as per the code instruction.

9. **Why is `delay(100)` used in the code?**

Answer: `delay(100)` introduces a short pause (100 milliseconds) to allow the servo to reach its position and to prevent rapid fluctuations in movement due to sensor noise.

10. **How can we modify the code to rotate the servo to 45° instead of 90°?**

Answer: We can change `Servopin.write(90);` in the code to `Servopin.write(45);` to rotate the servo to 45° when the IR sensor detects an object.

11. **If the servo motor is not moving even when the IR sensor detects an object, what could be the possible reasons?**

Answer: Possible reasons include incorrect wiring of servo power and signal pins, insufficient power supply, incorrect Arduino pin assignment in the code, or a faulty servo motor.

12. **How can we reduce false triggering of the IR sensor in bright sunlight?**

Answer: We can use an IR sensor with modulated IR signals, place the sensor in a shaded enclosure, or use shielding to minimize interference from ambient light.

13. **In real-world applications, where can we use this IR sensor–servo motor combination?**

Answer: It can be used in automatic doors, robotic arms for object sorting, obstacle detection and avoidance systems, and automated waste segregation systems.

14. **How would you modify this system to detect objects at multiple angles?**

Answer: By mounting the IR sensor on the servo motor and rotating it periodically, we can scan multiple angles for object detection, creating a simple radar-like system.

15. **If we want the servo to return to its initial position only after 5 seconds of no object detection, how can we modify the code?**

Answer: We can use the `millis()` function to measure the time since the last object detection and add a conditional check to move the servo back to its original position only after 5000 milliseconds have passed.

EXPERIMENT 7
BLINK LED USING ESP32.
VIVA QUESTIONS

1. What is the ESP32 and who developed it?

Answer: ESP32 is a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth, developed by Espressif Systems.

2. What is the operating voltage of ESP32 GPIO pins?

Answer: The GPIO pins of ESP32 operate at 3.3V logic level.

3. Why do we connect a resistor in series with an LED?

Answer: A resistor limits the current flowing through the LED, preventing it from burning out.

4. Which GPIO pin is used in this experiment for LED blinking?

Answer: GPIO 18 is used to connect the LED in this experiment.

5. What is the function of the `pinMode()` function in Arduino IDE?

Answer: `pinMode()` is used to configure a specific GPIO pin as either input or output.

6. What does the `digitalWrite()` function do in Arduino IDE?

Answer: `digitalWrite()` sets a pin's output level to HIGH (3.3V) or LOW (0V).

7. Why do we use the `delay()` function in the program?

Answer: The `delay()` function introduces a pause (in milliseconds) between instructions, which is used here to control the LED blinking interval.

8. What is the difference between ESP32 and Arduino UNO in terms of connectivity?

Answer: Unlike Arduino UNO, the ESP32 has built-in Wi-Fi and Bluetooth, making it more suitable for IoT applications.

9. What does the term IoT (Internet of Things) mean?

Answer: IoT refers to a network of interconnected devices that communicate and exchange data over the internet, often without human intervention.

10. What precautions should be taken while working with ESP32 circuits?

Answer: Avoid working with wet hands, keep the work area dry, always connect components properly, and disconnect power if overheating or burning smell is noticed.

EXPERIMENT 8

LDR INTERFACING WITH ESP32.

VIVA QUESTIONS

1. **What is the full form of LDR?**

Answer: LDR stands for Light Dependent Resistor, also known as a photoresistor or photocell.

2. **What is the working principle of an LDR?**

Answer: The working principle of an LDR is photoconductivity. Its resistance decreases as the intensity of incident light increases.

3. **What are the typical resistance values of an LDR in light and dark conditions?**

Answer: In darkness, resistance is very high (in the range of 1 MΩ to 10 MΩ), and in bright light, it drops to a few hundred ohms.

4. **What is the role of the resistor in the LDR module?**

Answer: The resistor forms a voltage divider circuit with the LDR, converting the change in resistance into a measurable voltage signal for the microcontroller.

5. **Why do we use a comparator (like LM393) in some LDR modules?**

Answer: A comparator is used to convert the analog voltage from the LDR into a digital HIGH/LOW signal based on a set threshold.

6. **Which pins of ESP32 are connected to the LDR module in this experiment?**

Answer: The D0 pin of the LDR module is connected to GPIO5 of ESP32, and the LED is connected to GPIO18 through a 220 Ω resistor.

7. **Why is a 220 Ω resistor used with the LED in this experiment?**

Answer: The resistor limits the current flowing through the LED to prevent damage. Without it, the LED may burn out due to excess current.

8. **What are the two types of outputs provided by an LDR module?**

Answer: The two outputs are:

- **Analog Output (A0):** Continuous voltage proportional to light intensity.
- **Digital Output (D0):** HIGH or LOW signal depending on threshold.

9. **Mention two real-life applications of LDR.**

Answer:

- Automatic street lighting (lights ON at night, OFF in daytime).
- Solar tracking systems to adjust solar panels based on sunlight direction.

10. **What will happen to the LED in this experiment if the LDR is placed in complete darkness?**

Answer: In darkness, the LDR module output will be LOW, so the ESP32 will turn OFF the LED.

EXPERIMENT 11

DESIGN AND 3D PRINT A ROCKET

VIVA QUESTIONS

1. **What is the purpose of using TinkerCAD in this experiment?**

Answer: TinkerCAD is used for creating the 3D model of the rocket using basic geometric shapes and preparing it for 3D printing.

2. **Why do we export the model in STL format?**

Answer: STL format is widely used in 3D printing because it stores the surface geometry of the model in a tessellated form that slicing software can process.

3. **What is the role of slicing software in 3D printing?**

Answer: Slicing software converts the 3D model into layers and generates G-code, which provides instructions to the 3D printer on how to build the object.

4. **Why do we align and group objects in TinkerCAD before exporting?**

Answer: Aligning and grouping ensures that all parts of the design fit together properly and behave as a single object during export and printing.

5. **What does “layer-by-layer” manufacturing mean in 3D printing?**

Answer: It means the printer builds the object one thin horizontal slice at a time, stacking layers to form the complete 3D shape.

6. **Why is PLA often chosen as a filament for educational projects?**

Answer: PLA is biodegradable, easy to print with, requires lower temperatures, and produces less warping, making it ideal for beginners.

7. **What precaution should be taken regarding the heated bed and nozzle of a 3D printer?**

Answer: They become very hot during operation, so direct contact should be avoided to prevent burns.

8. **What is the advantage of using 3D printing over traditional manufacturing for this project?**

Answer: 3D printing allows quick prototyping, customization of design, and fabrication without the need for molds or complex machinery.

9. **Why is it important to check the dimensions of the model before printing?**

Answer: Ensuring correct dimensions prevents misfits, wasted material, and ensures the printed rocket matches the intended design size.

10. **What could be a common issue if the 3D print does not stick to the build plate?**

Answer: The bed might not be levelled properly, or the first layer settings in the slicer may need adjustment for better adhesion.

EXPERIMENT 12

BUILD A LIVE SOIL MOISTURE MONITORING PROJECT, AND MONITOR SOIL MOISTURE LEVELS OF A REMOTE PLAN IN YOUR COMPUTER DASHBOARD

VIVA QUESTIONS

1. **What is the purpose of a soil moisture sensor in this experiment?**

Answer: The soil moisture sensor is used to measure the water content in the soil so that the plant's watering needs can be monitored in real time and displayed on the Blynk dashboard.

2. **Why do we use a capacitive soil moisture sensor instead of a resistive one?**

Answer: Capacitive sensors are preferred because they are less prone to corrosion, provide more stable readings, and have a longer lifespan compared to resistive sensors.

3. **Which microcontroller is used in this experiment and why?**

Answer: The ESP32 development board is used because it has built-in Wi-Fi, multiple ADC pins, and enough processing power to read sensor values and send them to the Blynk Cloud.

4. **What role does the ADC (Analog-to-Digital Converter) play in this project?**

Answer: The ADC in the ESP32 converts the analog voltage output from the soil moisture sensor into a digital value that can be processed and transmitted to the dashboard.

5. **What is the function of the Blynk platform in this experiment?**

Answer: Blynk is an IoT platform that allows real-time monitoring and control of devices. Here, it is used to create a web dashboard that displays soil moisture levels remotely.

6. **Why do we use a virtual pin (V0) in Blynk instead of a physical pin?**

Answer: Virtual pins in Blynk are software-based data channels that allow flexible communication between the hardware and the cloud dashboard, independent of actual GPIO hardware pins.

7. **How does the ESP32 connect to the Blynk Cloud?**

Answer: The ESP32 connects to the Blynk Cloud using Wi-Fi credentials (SSID and password) along with the Blynk Auth Token, which securely links the device to the user's cloud account.

8. **What precautions should be taken when using the soil moisture sensor?**

Answer: The sensor should not be powered continuously to prevent corrosion (especially resistive types), it should be inserted carefully into the soil near the plant roots, and electrical connections should be kept dry.

9. **What are some real-world applications of soil moisture monitoring systems?**

Answer: Applications include smart irrigation systems, greenhouse automation, precision agriculture, water conservation in urban gardens, and environmental monitoring projects.

10. **What are the limitations of low-cost soil moisture sensors used in this experiment?**

Answer: Limitations include dependence on soil type and salinity, reduced accuracy compared to industrial-grade sensors, the need for calibration, and possible degradation over time.